Lecture 10. Splines and Generalized Additive Models

Polynomial regression, regression and smoothing splines, local regression, GAMs

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10.1 Beyond linearity

Motivation. Relax linearity step by step while keeping interpretability.

Polynomials and step functions, regression and smoothing splines, local regression, GAMs. Example: `ISLR2::Wage`, 3000 male workers in the US Mid-Atlantic.

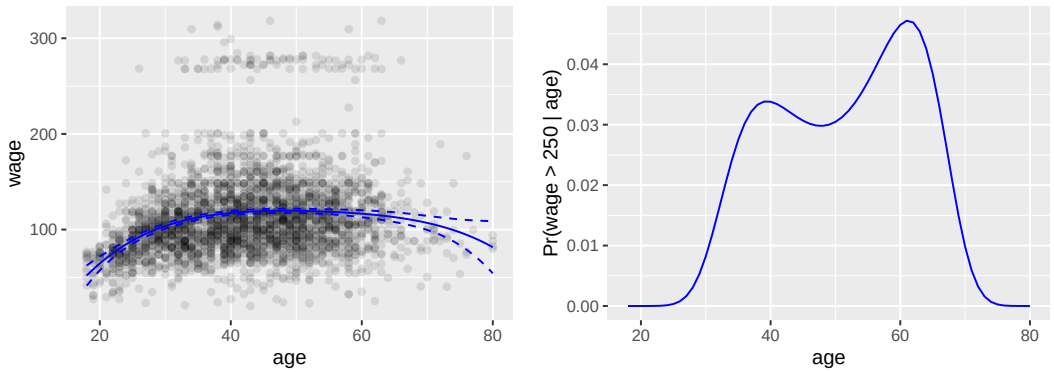
10.2 Polynomial regression and step functions

Polynomial regression replaces the line by

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \epsilon_i,$$

linear in the coefficients (least squares). Degrees above 4 behave strangely near the boundary. We look at $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \cdots + \hat{\beta}_4 x_0^4$ with a ± 2 SE band, not the coefficients.

```
fit4 <- lm(wage ~ poly(age, 4), data = Wage)
grid <- tibble(age = 18:80)
pr <- predict(fit4, grid, se.fit = TRUE)
grid <- grid %>% mutate(fit = pr$fit, lo = fit - 2 * pr$se.fit, hi = fit + 2 * pr$se.fit
    )
ggplot(Wage, aes(age, wage)) + geom_point(alpha = 0.1) +
  geom_line(data = grid, aes(y = fit), colour = "blue") +
  geom_line(data = grid, aes(y = lo), colour = "blue", linetype = "dashed") +
  geom_line(data = grid, aes(y = hi), colour = "blue", linetype = "dashed")
fit_log <- glm(I(wage > 250) ~ poly(age, 4), family = binomial, data = Wage)
grid %>% mutate(p = predict(fit_log, grid, type = "response")) %>%
  ggplot(aes(age, p)) + geom_line(colour = "blue") + labs(y = "Pr(wage > 250 | age)")
```



A high-earner group (above \$250 000) stands apart; logistic regression for $\Pr(\text{wage} > 250)$:

$$P(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)};$$

its intervals (not drawn) are wide: few high earners.

Step functions avoid global structure: cut X at $c_1 < \cdots < c_K$ into $K+1$ bins,

$$C_0(X) = I(X < c_1),\ C_1(X) = I(c_1 \le X < c_2),\ \ldots,\ C_K(X) = I(c_K \le X),$$

summing to 1. In $y_i = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i) + \epsilon_i$, β_0 is the mean for $X < c_1$ and $\beta_0 + \beta_j$ the mean in $[c_j, c_{j+1})$.

```
coef(lm(wage ~ cut(age, 4), data = Wage))
```

```
##           (Intercept)   cut(age, 4)(33.5,49]   cut(age, 4)(49,64.5]
##             94.158392              24.053491              23.664559
## cut(age, 4)(64.5,80.1]
##              7.640592
```

They miss within-bin trends.

10.3 Regression splines

Motivation. Splines are local like step functions and smooth like polynomials.

Basis functions. Polynomial and step regression are special cases of

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \epsilon_i$$

with fixed basis functions ($b_j(x) = x^j$; $b_j(x) = I(c_j \le x < c_{j+1})$): a linear model in $b_j(x_i)$.

Piecewise polynomials. Coefficients change at **knots**. One knot at c:

$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \epsilon_i, & x_i < c, \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \epsilon_i, & x_i \ge c. \end{cases}$$

Requiring continuity and continuous first and second derivatives at c gives a **cubic spline**: the right piece keeps 1 of its 4 degrees of freedom. K knots: $K + 4$ degrees of freedom.

Spline basis. $y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \epsilon_i$: x, x^2, x^3 plus one **truncated power function** per knot,

$$h(x, \xi) = (x - \xi)_+^3 = \begin{cases} (x - \xi)^3, & x > \xi, \\ 0, & \text{otherwise.} \end{cases}$$

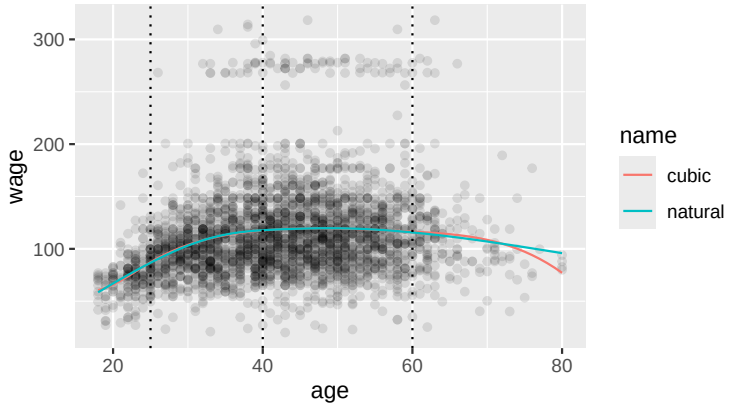
which breaks only the third derivative at ξ: $K + 4$ coefficients.

Why the truncated power basis works [Supplement]

$h(x, \xi)$ and its first two derivatives, $3(x - \xi)_+^2$ and $6(x - \xi)_+$, vanish at $x = \xi$ from both sides, while the third derivative jumps from 0 to 6. Conversely, any cubic spline with knots $\xi_1 < \cdots < \xi_K$ differs on $[\xi_k, \xi_{k+1})$ from its expression on $[\xi_{k-1}, \xi_k)$ by a cubic that vanishes with two derivatives at ξ_k, which must be $c\,(x - \xi_k)^3$. So the $K + 4$ functions span exactly the cubic splines.

`bs()` gives a stabler basis (B-splines) of the same space:

```
fit_bs <- lm(wage ~ bs(age, knots = c(25, 40, 60)), data = Wage)
fit_ns <- lm(wage ~ ns(age, df = 4), data = Wage)
grid <- tibble(age = 18:80) %>%
  mutate(cubic = predict(fit_bs, .), natural = predict(fit_ns, .))
ggplot(Wage, aes(age, wage)) + geom_point(alpha = 0.1) +
  geom_line(data = pivot_longer(grid, -age), aes(y = value, colour = name)) +
  geom_vline(xintercept = c(25, 40, 60), linetype = "dotted")
```



Natural splines (`ns()`) are linear beyond the boundary knots, taming the high variance at the ends.

Knots. Usually at quantiles of X (`ns(age, df = 4)`); their number by cross-validation:

```
set.seed(1)
folds <- sample(rep(1:10, length.out = nrow(Wage)))
cv_ns <- function(df) mean(map_dbl(1:10, function(k) {
  fit <- lm(wage ~ ns(age, df = df), data = Wage[folds != k, ])
  mean((Wage$wage[folds == k] - predict(fit, Wage[folds == k, ]))^2)
}))
round(map_dbl(1:8, cv_ns), 1)
```

```
## [1] 1676.8 1602.4 1595.4 1595.2 1595.1 1594.0 1593.9 1593.6
```

CV error drops up to about 3 df, then is flat. Natural splines beat equal-df polynomials at the boundaries (knots, not high powers).

10.4 Smoothing splines

Motivation. A knot at every point, with a roughness penalty instead of knot choice.

An unconstrained g interpolates (RSS $= 0$). The **smoothing spline** minimises

$$\sum_{i=1}^{n} \big(y_i - g(x_i)\big)^2 + \lambda \int g''(t)^2 \, dt, \qquad \lambda \geq 0.$$

"Loss + penalty": $\int g''^2$ measures roughness. $\lambda = 0$ interpolates; $\lambda \to \infty$ gives a line. The minimiser is a shrunken natural cubic spline with knots at the unique x_i.

Effective degrees of freedom. With $\hat{g}_\lambda = S_\lambda y$ ($n \times n$ smoother matrix),

$$\mathrm{df}_\lambda = \sum_{i=1}^{n} \{S_\lambda\}_{ii},$$

decreasing from n to 2 (Exercise 10.3); for least squares it is the number of parameters [Supplement].

Choosing λ. LOOCV from one fit:

$$\mathrm{RSS}_{\mathrm{cv}}(\lambda) = \sum_{i=1}^{n} \big(y_i - \hat{g}_\lambda^{(-i)}(x_i)\big)^2 = \sum_{i=1}^{n} \left[\frac{y_i - \hat{g}_\lambda(x_i)}{1 - \{S_\lambda\}_{ii}}\right]^2,$$

with $\hat{g}_\lambda^{(-i)}$ fitted without observation i.

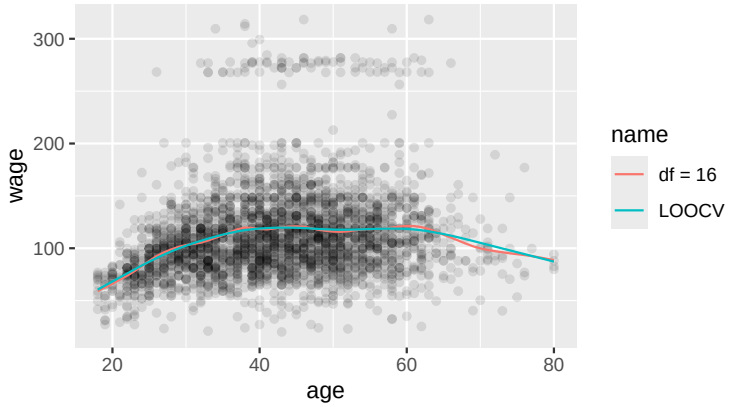
```
ss16 <- smooth.spline(Wage$age, Wage$wage, df = 16)
ss_cv <- smooth.spline(Wage$age, Wage$wage, cv = TRUE)
```

```
## Warning in smooth.spline(Wage$age, Wage$wage, cv = TRUE): cross-validation with non-unique 'x'
##     values seems doubtful
```

```
c(df_chosen = ss_cv$df, lambda = ss_cv$lambda)
```

```
##  df_chosen     lambda
## 6.79459570 0.02792303
```

```
tibble(age = 18:80, `df = 16` = predict(ss16, 18:80)$y,
       `LOOCV` = predict(ss_cv, 18:80)$y) %>%
  pivot_longer(-age) %>%
  ggplot(aes(age, value, colour = name)) +
  geom_point(data = Wage, aes(age, wage), inherit.aes = FALSE, alpha = 0.1) +
  geom_line() + labs(y = "wage")
```

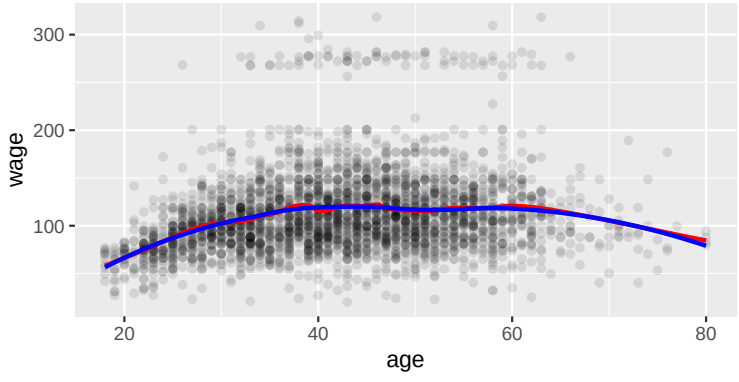


LOOCV (about 7 df) is smoother than 16 df and fits almost as well.

10.5 Local regression

LOESS (Lecture 2) fits at each x_0 a weighted line on the nearest fraction s (**span**) of points; smaller span, wigglier fit. It is memory-based and poor beyond three or four dimensions.

```
ggplot(Wage, aes(age, wage)) + geom_point(alpha = 0.1) +
  geom_smooth(method = "loess", span = 0.2, se = FALSE, colour = "red", formula = y ~ x)
    +
  geom_smooth(method = "loess", span = 0.5, se = FALSE, colour = "blue", formula = y ~ x
    )
```



10.6 Generalized additive models

Motivation. GAMs add one smooth function per predictor.

A GAM replaces each $\beta_j x_{ij}$ by a smooth function:

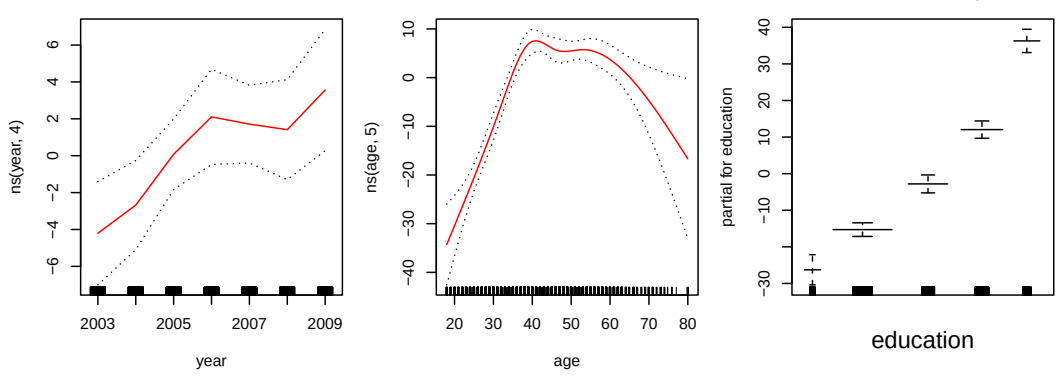
$$y_i = \beta_0 + \sum_{j=1}^{p} f_j(x_{ij}) + \epsilon_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \epsilon_i.$$

Additive: separate f_j per X_j, built from the smoothers above. For the wage data,

$$\text{wage} = \beta_0 + f_1(\text{year}) + f_2(\text{age}) + f_3(\text{education}) + \epsilon,$$

with natural splines for year and age and dummies for education: one big regression, fitted by `lm()`:

```
gam1 <- lm(wage ~ ns(year, 4) + ns(age, 5) + education, data = Wage)
par(mfrow = c(1, 3), mar = c(4, 4, 1, 1))
plot.Gam(gam1, se = TRUE, col = "red")
```



Others fixed, wage rises slightly with year, peaks in middle age, rises with education. Smoothing splines: `gam::gam(wage ~ s(year, 4) + s(age, 5) + education)` (backfitting).

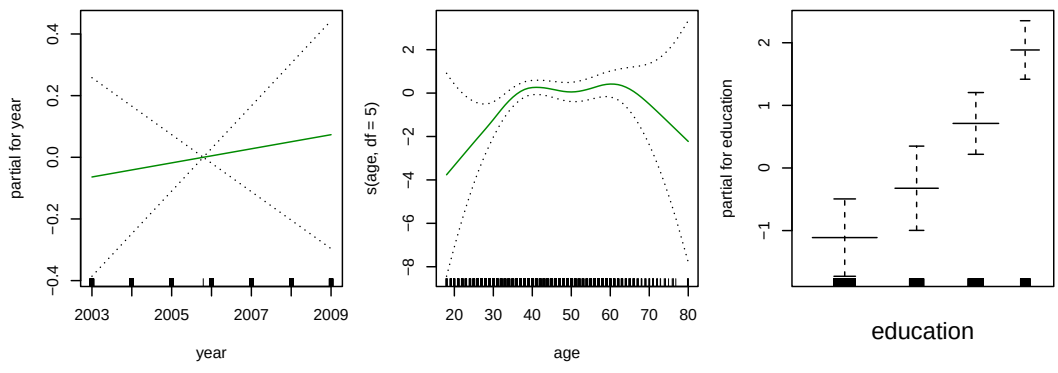
Pros. Automatic nonlinear f_j, often better prediction, per-variable effects, smoothness summarised by df. **Cons.** Additivity misses interactions; add $X_j \times X_k$ or $f_{jk}(X_j, X_k)$ by hand, or use random forests and boosting.

GAMs for classification. For binary Y with $p(X) = P(Y = 1 \mid X)$, the logistic GAM is

$$\log \frac{p(X)}{1 - p(X)} = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p).$$

For wage above \$250 000:

```
gam_lr <- gam(I(wage > 250) ~ year + s(age, df = 5) + education,
              family = binomial, data = Wage, subset = (education != "1. < HS Grad"))
par(mfrow = c(1, 3), mar = c(4, 4, 1, 1))
plot(gam_lr, se = TRUE, col = "green4")
```



The lowest education level has no high earners and is excluded (separation, Lecture 9) [Supplement].

Summary. Regression splines: basis functions; smoothing splines: loss plus penalty; GAMs: $x_j \to f_j(x_j)$, additively.

10.7 Selected exercises

Motivation. Exact degrees of freedom of the lecture's smoothers.

Exercise 10.1 (dimension of spline spaces)

Let $\xi_1 < \cdots < \xi_K$ be knots.
(a) Show that the functions $1, x, x^2, x^3, (x-\xi_1)_+^3, \ldots, (x-\xi_K)_+^3$ are linearly independent, so that cubic splines with these knots form a space of dimension $K+4$.
(b) A *natural* cubic spline is in addition linear on $(-\infty, \xi_1]$ and on $[\xi_K, \infty)$. For $K \geq 2$, show that natural cubic splines form a space of dimension K. (This is why `ns(age, df = 4)`, which uses three interior and two boundary knots, has four columns besides the intercept.)

Solution [Supplement].

(a) Suppose $f(x) = \sum_{j=0}^{3} c_j x^j + \sum_{k=1}^{K} \theta_k (x-\xi_k)_+^3 = 0$ for all x. On $(-\infty, \xi_1)$ all truncated terms vanish, so a cubic polynomial is zero on an interval and $c_0 = \cdots = c_3 = 0$. Inductively, if $\theta_1 = \cdots = \theta_{k-1} = 0$, then on (ξ_k, ξ_{k+1}) (with $\xi_{K+1} = \infty$) $f(x) = \theta_k (x-\xi_k)^3 = 0$, so $\theta_k = 0$. The notes showed that these functions span the cubic splines, so they form a basis.

(b) Write a cubic spline in the basis of (a).

Step 1. On $(-\infty, \xi_1)$, $f = c_0 + c_1 x + c_2 x^2 + c_3 x^3$, which is linear iff $c_2 = c_3 = 0$.

Step 2. On (ξ_K, ∞) every truncated term is active, and the coefficients of x^3 and x^2 are $c_3 + \sum_k \theta_k$ and $c_2 - 3\sum_k \theta_k \xi_k$. Given $c_2 = c_3 = 0$, linearity there means $\sum_k \theta_k = 0$ and $\sum_k \theta_k \xi_k = 0$.

Step 3. The four constraints are linear. The first two involve only c_2, c_3; the last two involve only θ, with coefficient vectors $(1, \ldots, 1)$ and $(\xi_1, \ldots, \xi_K)$, which are linearly independent because the knots are distinct and $K \geq 2$. So the constraints are independent and the dimension is $(K+4) - 4 = K$. ∎

Exercise 10.2 (degrees of freedom of a step function)

Let every bin of the step-function regression of Section 10.2 contain at least one observation, and let n_j be the size of bin $j = 0, \ldots, K$. Show that the hat matrix H (with $\hat{y} = Hy$) replaces each y_i by the mean of its bin, and that $\mathrm{tr} H = K + 1$, the number of coefficients. Why must C_0 be left out when an intercept is included?

Solution [Supplement].

Step 1. Since $C_0 + \cdots + C_K \equiv 1$, the columns $1, C_0, \ldots, C_K$ are linearly dependent and least squares would not have unique coefficients; dropping C_0 leaves $K+1$ independent columns spanning the same space as $C_0, \ldots, C_K$.

Step 2. The fitted function is constant on each bin, with value $a_0 = \beta_0$ on bin 0 and $a_j = \beta_0 + \beta_j$ on bin $j \geq 1$ (a bijection between the a's and the β's), and $\mathrm{RSS} = \sum_j \sum_{i \in \text{bin } j} (y_i - a_j)^2$ separates over bins. By the constant-model result of Lecture 5 each a_j is minimised by the bin mean $\bar{y}_{(j)}$.

Step 3. Ordering the observations by bin, H is block diagonal with blocks $n_j^{-1} \mathbf{1}\mathbf{1}^{\mathsf{T}}$, the matrix that averages within the bin. Each block has trace $n_j \cdot n_j^{-1} = 1$, so $\mathrm{tr} H = K + 1$. This agrees with the definition $\mathrm{df} = \mathrm{tr} S$ used for smoothing splines. ∎

Exercise 10.3 (effective degrees of freedom decrease with λ)

With $n \geq 3$ distinct values x_i, the smoothing-spline fitted values can be written $\hat{g}_\lambda = S_\lambda y$ with $S_\lambda = (I + \lambda Q)^{-1}$, where Q is a fixed symmetric positive semidefinite $n \times n$ matrix whose null space consists of the vectors $(a + bx_i)_{i=1}^n$ (a standard fact from the theory of smoothing splines; take it as given). Show that

$$\mathrm{df}_\lambda = \mathrm{tr} S_\lambda = \sum_{k=1}^{n} \frac{1}{1 + \lambda d_k},$$

where $d_k \geq 0$ are the eigenvalues of Q, that df_λ is strictly decreasing in λ, and that it tends to n as $\lambda \to 0$ and to 2 as $\lambda \to \infty$.

Solution [Supplement].

Step 1. Write $Q = U \,\mathrm{diag}(d_1, \ldots, d_n) U^{\mathsf{T}}$ with U orthogonal. Then $I + \lambda Q = U \,\mathrm{diag}(1 + \lambda d_k) U^{\mathsf{T}}$ has positive eigenvalues, and $S_\lambda = U \,\mathrm{diag}\{1/(1 + \lambda d_k)\} U^{\mathsf{T}}$. The trace is invariant under orthogonal similarity, so $\mathrm{df}_\lambda = \sum_k 1/(1 + \lambda d_k)$.

Step 2. The null space of Q is spanned by $(1, \ldots, 1)$ and $(x_1, \ldots, x_n)$, which are independent since the x_i are not all equal. So exactly two eigenvalues are 0 and $n - 2 \geq 1$ are positive.

Step 3. $\frac{d}{d\lambda} \mathrm{df}_\lambda = -\sum_k d_k/(1 + \lambda d_k)^2 < 0$, since some $d_k > 0$. As $\lambda \to 0$ every term tends to 1; as $\lambda \to \infty$ the terms with $d_k > 0$ tend to 0 and the two with $d_k = 0$ stay 1.

Step 4. The eigenvalues $1/(1 + \lambda d_k) \in (0, 1]$: linear trends ($d_k = 0$) are kept, rough components (large d_k) shrunk most, as in ridge. ■